

Siddharth Potta

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EDUCATION

University of Texas at Austin

Bachelor of Science, Computer Science

• GPA: 3.82

• Coursework: Data Structures & Algorithms, Computer Architecture, Operating Systems, Advanced Computer Architecture (In progress), Algorithms

May 2024 - 2027

EXPERIENCE

Intel

System Software Engineer Intern

GuC team (GPU firmware)

- Collaborated with Microsoft, Linux, and graphics teams to develop a Python-based graphics micro-controller performance analysis tool used across multiple Intel GPU teams to debug firmware behavior, analyze performance bottlenecks, and accelerate validation workflows.
- Optimized the GPU microkernel scheduler, reducing context-switch overhead by improving frame-submission throughput under heavy loads and enhancing gaming performance.
- Hardened GPU firmware by tuning the microkernel scheduler, resolving critical race conditions, and enriching crash-dump output with scheduler and register state to enable faster live-system debugging and root-cause analysis.

Apr 2026 - Present

Folsom California

Longhorn Developers

Lead Developer

- Architected and currently lead a team of 6 engineers building UT Degree Audit Plus, a Chrome extension serving 50,000+ UT students by automating course planning and registration.
- Built and tested a RAG-based system with Llama 3 that transforms unstructured degree requirement data into personalized academic plans by retrieving and reasoning over student preferences and degree requirements.
- Led quality assurance for the Chrome extension by using Playwright for automated testing, establishing bug-reporting workflows, and conducting code reviews to verify data accuracy against UT Registrar systems.
- Stack: React, WXT, Typescript, PostgreSQL, IndexedDB, Ollama.

Aug 2025 - Present

Austin, TX

Texas Convergent

Software Developer

- Led a team of 7 in developing a proprietary AI Dance Coach for local dance teams and studios, leveraging computer vision to deliver real-time feedback during training sessions.
- Architected a data-intensive pose-estimation pipeline using MediaPipe and Python, incorporating vector analysis and Euclidean distance algorithms to quantify dancer accuracy.
- Designed an interactive feedback system with real-time visual overlays, applying software engineering best practices to enhance training sessions.

Jan 2025 - May 2025

Austin, TX

ACTIVITIES

Texas Convergent

Build Team Mentor

- Hosting weekly workshops teaching Machine Learning, Full Stack software development, and powerful tools such as PyTorch, Google Cloud, Supabase, React JS.
- Provide technical mentorship to 21 students across 4 project teams, offering guidance on project architecture and agile methodologies to build and deploy public, user-centric applications.

May 2024 - Present

PROJECTS

End to End LLM Inference Accelerator

- Built a modular, reusable AI inference accelerator that runs multiple transformer architectures end-to-end, including Mixture-of-Experts (MoE), multi-token prediction (MTP), and diffusion text models, backed by a custom C++ inference engine, an op-graph IR with INT8 quantization and kernel fusion, and pluggable hardware compute backends.
- Wrote custom GPU compute kernels (CUDA: naive → shared-memory tiling → tensor cores) and a SystemVerilog systolic-array matrix-multiply core, exposed through a unified backend interface that runs the same model on CPU, GPU, and custom RTL for head-to-head benchmarking - achieving $[X] \times$ matmul throughput and $[Y]$ tokens/sec.
- Architected for horizontal and vertical scaling: pluggable systolic-array (TPU-style) and streaming-dataflow (Groq-LPU-style) compute backends plus drop-in optimizer passes **deployed live on FPGA as an on-device inference agent**.

OOO CPU + Advanced Memory Subsystem <https://github.com/Sidd03192/OutOfOrder-CPU>

- Designed and taped out a fully pipelined out-of-order (OOO) processor in SystemVerilog on an Intel FPGA, featuring a multi-level cache hierarchy (L1/L2), hardware TLB, and Load-Store Queue - synthesized and validated on real silicon via Quartus.
- Engineered an automated co-simulation verification framework pairing Verilator RTL simulation with a golden C ISS, diffing architectural state across AArch64 ELF binaries to diagnose microarchitectural bugs that unit tests alone could not detect.
- Extended the AArch64 ISA with full exception/syscall support (SVC) and hardware low-power mode, differentiating fault vs. clean-termination paths at the EL1 exception handler level and verified end-to-end on physical FPGA - confirmed functional by thermal signature.

Codesprout.net www.codesprout.net

- Built a full-stack EdTech LMS that automates coding assignment generation, grading, and lesson plan generation using fine-tuned Ai Agents, cutting grading time by 90% for 5 pilot users..
- Deployed scalable backend infrastructure using Google Cloud Run and Docker to handle concurrent auto-grading requests, Kafka to handle multiple AI pipelines & Redis for caching.
- Architected production-grade agentic AI systems using LangChain, including Code-Act agents for reliable tool execution and grading, agentic RAG pipelines for personalized student learning, and React-style agent orchestration for multi-step assignment and curriculum generation for educators.

x86 Mini OS

- Implemented core OS subsystems in C for a lightweight x86-based operating system (Pintos), including kernel threads, priority-based scheduling, and user program support, with a strong focus on synchronization and race-free concurrency.
- Implemented a Linux-style file system and virtual memory system with demand paging, reducing swap access, enhancing on-disk Inode structures, directory/path resolution, and buffer caching to improve throughput and reduce disk I/O latency.
- Collaborated in a 3 person engineering team to design, implement, and debug low-level OS features in C (system calls, process management, and file system primitives) within an x86 emulator environment (bochs), using formal design docs, code reviews, and systematic test harnesses.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C++, JavaScript/TypeScript, SQL, Assembly, C, Rust, Go
- **Frameworks:** Next.js, Spring Boot, Flask, Node.js, Tailwind CSS, Fast api, React, Playwright, Redux, ORMs (Drizzle)
- **Cloud & DevOps:** Docker, Kubernetes, Google Cloud Platform (GCP), Azure, Git, CI/CD, Distributed Systems, AWS
- **Data & Engineering Practices:** Scikit-Learn, MediaPipe, OpenAI API, LangChain, RAG Pipelines, Pandas, Modal, PostgreSQL, Software Engineering Best Practices, Agile Development, API Design